www.tntextbooks.in

is the most abundant Carotene in plants and it is a precursor of Vitamin A. Lycopene is the red pigment found in the fruits of tomato, red peppers and roses.

ii. Xanthophylls:

Yellow ($C_{40}H_{56}O_2$) pigments are like carotenes but contain oxygen. Lutein is responsible for yellow colour change of leaves during autumn season. Examples: Lutein, Violaxanthin and Fucoxanthin.

13.3.3 Phycobilins

They are proteinaceous pigments, soluble in water, and do not contain Mg and Phytol tail. They exist in two forms such as 1. Phycocyanin found in cyanobacteria 2. Phycoerythrin found in rhodophycean algae (Red algae).

13.4 Spectrum of Electromagnetic Radiation

In the total electromagnetic spectrum,visible light is the smallest part. The entire life on earth depends on light and is the driving force for all organisms. Plants have natural potential to utilize solar energy directly. In the given picture electromagnetic radiation spectrum and components of visible spectrum are mentioned. The wavelength of solar radiation which reaches the earth is between 300 to 2600 nm. The visible spectrum ranges between 390 to 763 nm (3900 Å to 7630 Å). The colour of the light is determined by the wavelength. Energy of the quantum is inversely proportional to wavelength. Shorter wavelength has

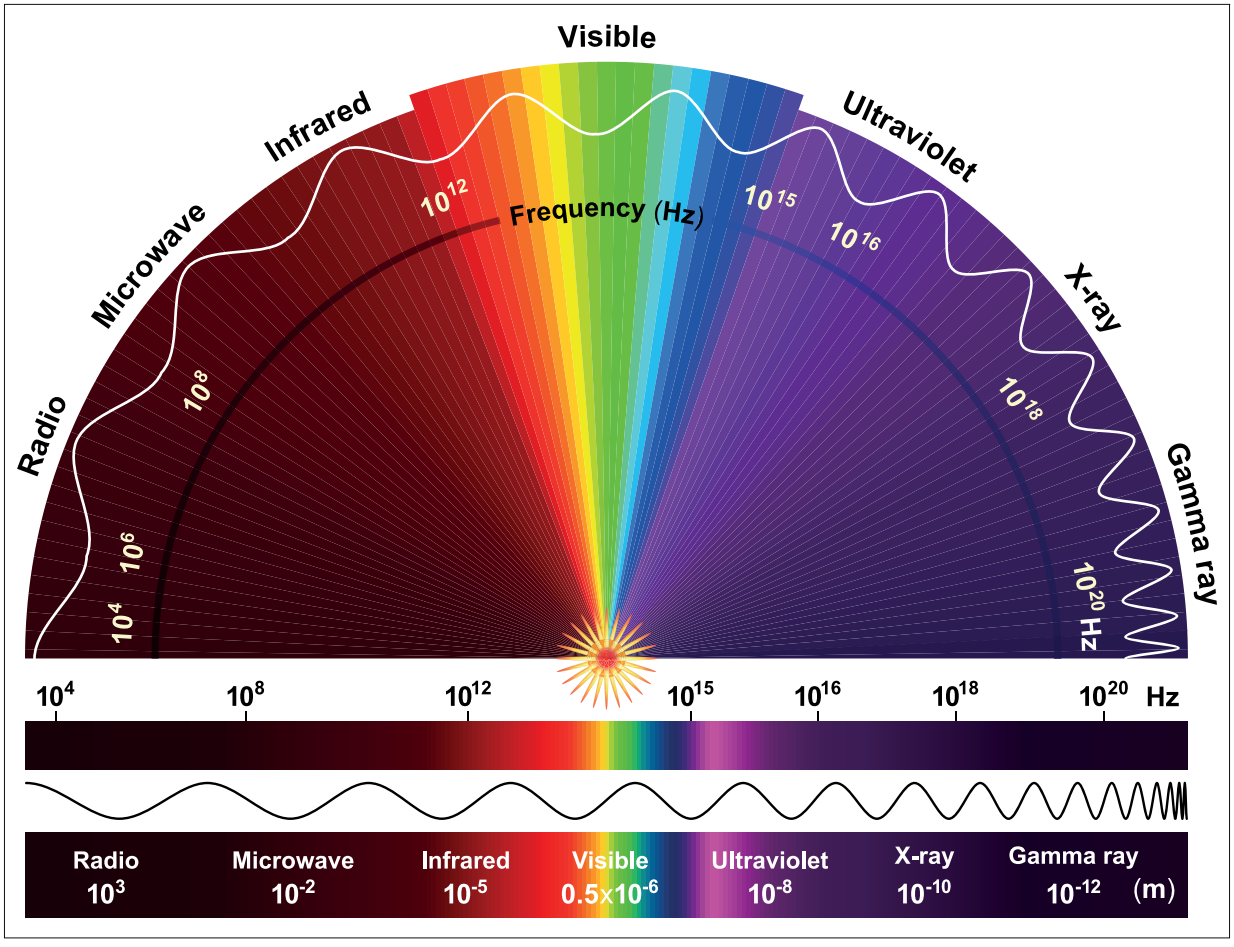


Figure 13.5: Electromagnetic Spectrum